

Wow! I Have Tiny Hands: Design And Evaluation Of Adaptive Virtual Hands For Small Object Selection Within Arms Length In Dense Virtual Environment

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ABSTRACT

Object selection is essential to VR applications, but selecting small objects within arms' reach in dense virtual environments can be challenging. We introduce Tiny Hands, which allows reducing the size of the dominant virtual hand in immersive VR to navigate efficiently and accurately select the desired object. This study compares the tiny hands technique with ray casting and pinch-to-select techniques. Our results indicate that the tiny hands technique is significantly faster and more accurate than the other two techniques. It is also significantly easier to use and learn, perceived as natural, playful, and the most preferred technique. Participants appreciated its ability to resize virtual hands for selecting small objects and ease in navigating a dense VE. They also liked its similarity to mouse and cursor-based interactions and flexibility in adjusting hand size according to various object sizes. Participants perceived tiny hands as potentially fatigued for prolonged use and reported a sense of incorrect depth perception when hand sizes are too small.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction techniques*; Gestural input.

KEYWORDS

Object selection, Small objects, Dense virtual environment, Selection technique

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1 INTRODUCTION

Virtual reality (VR) is a computer-generated simulation of a three-dimensional environment that provides users with a fully immersive and interactive experience, allowing them to feel like they are in a digitally created environment. VR applications provide the ability to select and manipulate (such as move, rotate, scale, and transform) virtual objects, navigate the virtual environment (VE), and perform application-specific tasks [1]. Object selection is an essential aspect of VR applications because it allows users to interact with and manipulate virtual objects in the VE. Object selection in VR involves identifying and choosing a virtual object within a 3D environment. In VR, VEs contain objects of varying densities, sizes, and distances from the user. For example, VR applications designed for drug discovery molecular visualization enable scientists to interact with and analyze small-molecule compounds and protein structures that drive disease. In order to conduct scientific analysis and relational mapping of molecules, scientists require the ability to interact with small-sized molecules physically, even those as small as nail-sized, that are located at varying distances within a dense VE, including small virtual objects within arm's reach. However, selecting small objects placed at arm's length in a dense VE can be challenging and lead to the inaccurate selection, increased task completion times, and higher levels of fatigue and frustration [1]. Therefore, it is necessary to develop effective object selection techniques that enable accurate and quick selection, particularly for small objects placed arm's length within dense VE.

In this paper, we present the Tiny Hands technique for selecting small objects, including nail-sized objects, placed within arm's reach (50 cm) in dense VE. Tiny hands allow users to scale down the size of virtual hands to easily navigate and select a small target inside a dense VE. We compared the tiny hands technique with existing techniques, ray-casting and pinch-to-select in literature. Our results show that tiny hands were significantly faster and more accurate in selecting small objects within arm's reach in dense VE

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compared to the other two techniques. It was also significantly easy to use, learn, and natural. The remainder of the paper is organized as follows. Section 2 gives an overview of the related work. Section 3 contains a detailed description of the design aspects of the tiny hands technique. Section 3 provides details of the evaluation process, including the study procedure, participants, study setup, and data collection methods. The 4th section elucidates the study's outcomes, followed by Section 5, which discusses the findings and the limitations. Lastly, in Section 6, we conclude the paper.

2 RELATED WORK

One of the earliest techniques for small object selection is by increasing the width of a target proposed by Kabbash and Buxton [8]. They investigated the use of area cursors that has a large activation region for small target selection. However, a problem of ambiguity arises in dense VEs, as multiple targets could fall inside the area cursor at one time. Various other improvements in area cursors techniques have been designed [6, 12, 15] however these techniques may actually perform worse in selecting small targets in dense environments since small movements will cause the cursor to constantly resize. These visualization techniques may also increase visual clutter and further increase difficulty in locating the small target in dense VEs. Findlater et al., [5] proposed a technique that allows users to select an area of the screen and expand the objects into arcs that can be selected by clicking and crossing rather than pointing. These techniques offer higher completion time for small targets and were negatively impacted in dense environments.

The use of virtual hand techniques [1] in the context of VR HMD provides a more natural way of interaction. However, the projection of the virtual hand occludes the underlying targets and adds visual clutter to the user interface, making it difficult to select small targets in dense VEs. Delamare et al. [3], proposed Multi-Finger-Bubble which aims at cluster of targets with their palm to select small targets. However, use of additional visualization of multiple lines in a dense VE have a negative effect on the overall user performance. Piumsombon et al., [11] proposed mini-me, a scaled down version of life-size avatar for simple selection task and for remote collaboration in mixed reality. Schjerlund et al., presented Ninja hands [13], that maps one physical hand to many distributed targets for near and distant object selection. However, the influence of target density, distance and size for object selection have not been investigated in depth for all of the techniques.

3 THE TECHNIQUE: TINY HANDS

The tiny hands technique is designed to facilitate the direct selection of small targets within arms' reach in dense VEs by scaling down the size of virtual hands. This technique involves reducing the size of the dominant virtual hand in immersive VR to navigate efficiently and accurately select the desired object fig 1d. To activate tiny hands, the non-dominant hand palm is faced downwards and held in front of the user (in the camera view) for 2 seconds. The user then performs a vertical upward/downward movement to increase/decrease the size of the tiny hands displayed on the dominant right hand. The size of the virtual hand can be scaled down to 0.1 from a gain factor of 1, which is the actual virtual hand size. The vertical movement in the upwards and downward

directions increases or decreases the hand size, respectively. Once the desired virtual hand size is achieved, the user points and intersects (or touch) the desired object to confirm the selection. Tiny hands are deactivated after the user holds the non-dominant hand in front of the user again for 2 seconds. We believe that the tiny hands technique allows for an improved selection of small objects within the arm's reach of dense VEs by reducing the size of the dominant virtual hand, making it easier for users to navigate and select objects with precision and accuracy.

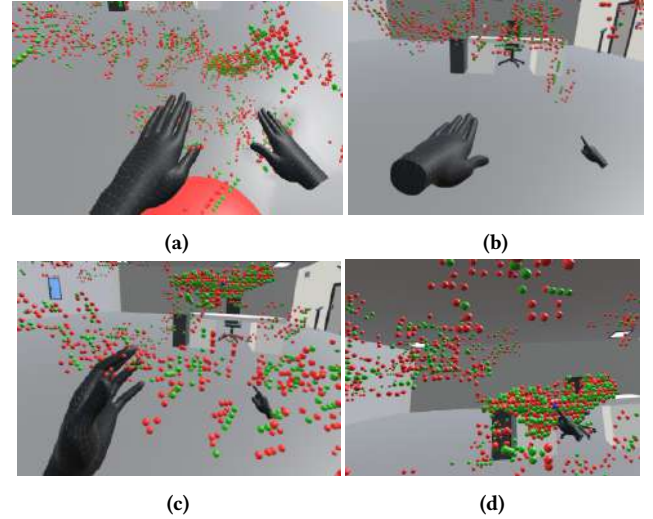


Figure 1: (a) The palm of the non-dominant is faced downwards and held for 2 seconds to trigger tiny hand on the dominant virtual hand (b) User then performs vertical downward movement to decrease the size of the tiny hands (c) the palm of the non-dominant hand facing downwards is removed from its position to start moving the tiny hand and (d) Using desired tiny hands size, the user accurately points and intersects (or touches) to select the small target.

4 EVALUATION STUDY

This evaluation study aims to investigate the effectiveness of the tiny hands technique compared to other chosen object selection techniques.

4.1 Baseline Techniques

To evaluate the effectiveness of the tiny hands technique, we compared it against two commonly used techniques in VR: ray casting [1] and the pinch-to-select [10] techniques. The ray casting technique involves pointing the index finger towards the target to cast a colored ray, which can be used to select the virtual. We adopted the ray casting technique because of its popularity, simplicity, and generality. It requires only two degrees of freedom and works at any distance [1]. On the other hand, the pinch-to-select technique is a popular and commonly used technique in commercially successful HMD devices, such as Oculus Quest, Microsoft HoloLens, etc. It uses a pinch metaphor for object selection, where the user



Figure 2: A participant using tiny hands technique. A video camera placed diagonally to the participant captured visual data for further analysis.

can control the size of the cursor by pinching their thumb and forefinger together. The cursor is displayed as a circular cone that can be squeezed or expanded based on the pinch gesture. Once the fingers are pinched together, the cursor acts like a laser pointer that can be used to make object selections. Additionally, this technique provides feedback on the direction in which the user is pointing.

4.2 Design of VE

For the study, we designed a VE with a dense molecular structure of 1192 spheres. The VE had dimensions of 6.5 m x 4.5 m x 3.5 m and was modeled after a small-molecule compound commonly used in molecular modeling by biologists to study potential effects on humans. The spheres in the structure were colored red and green and served as distractors, while the target objects were colored blue. The spheres were static and measured 0.4 mm in diameter each. The molecular structure was positioned 0.4 meters away from the participant.

4.3 Study Setup and Apparatus

The study used the Oculus Quest 1, and the Unity 3D game engine was utilized to develop the prototype. The visual stimulus was provided through the Oculus Quest HMD, which had a resolution of 1440 x 1600 per eye and a diagonal FOV of 115.52°. The Oculus Quest HMD was connected to a computer running on an i7 processor, integrated graphics 5500, 16 GB RAM, and a Microsoft Windows 10 operating system via the Oculus Link cable. During the experiment, the virtual hand (hand prefab provided by the Oculus Integration Kit) was scaled using the mouse scroll. A video camera was placed diagonally to the participant to capture visual data for further analysis of study results. Fig 2 shows the experimental setup used for the study.

4.4 Task

The task was to select the blue target placed at one location among the molecular structure. Each time the user selects the target, visual feedback in the form of changing the target color from blue to green was provided to indicate the selection, and the following target location was indicated. This was repeated for 5 target selections.

4.5 Participants

22 non-paid participants (12 males, 10 females) from 18-33 (Mean= 27.42, SD= 4.37) were chosen for the study. The participants were graduate students studying HCI. All participants had prior experience using HMD-VR platforms (e.g., Oculus Rift and/or HTC Vive). All these students had taken a VR course in the previous semester of their study program.

4.6 Study Procedure

The study was conducted in a controlled laboratory environment at the Indian Institute of Technology (IIT) Guwahati. Before the task, a moderator verbally introduced the study to each of the 22 participants. Participants were shown the VE and explained the three techniques being tested: the ray-casting technique, the pinch-to-select technique, and the tiny hands technique. The moderator trained each participant until they felt confident using each technique to complete the task. The training sessions lasted approximately 30 minutes per participant. To begin the study, each participant was randomly assigned one of the three techniques to complete the task, which was to select a blue target object. Participants were permitted to adjust the size of their tiny hands at any point during the study, and any changes to the size were also noted. After completing the task, post-task interviews were conducted to gain insight into the participant's overall perception of the proposed techniques. The entire study was video recorded and lasted approximately 90 minutes per participant.

4.7 Data Collection Method

The experiment involved 22 participants, who completed a total of 330 trials (3x5x22) using three techniques (Tiny hands, Raycasting, Pinch-to-select) to select 5 targets. During the experiment, we gathered user performance data using system logs and user preference data through questionnaires. Logs registered information regarding the task completion time and the number of incorrect selections. The task completion time was calculated after participants set the tiny hands size and pressed an OK button. An error was counted when the target selected by the user did not match the target specified during the task. The moderator noted the hand size selected by the participant. We provided a 7-point likert scale questionnaire to participants in a printed sheet to study ease of use and learning, naturalness, and playfulness. The questions were written as "this technique was easy to use", "this technique was easy to learn", "this technique was natural" and "this technique was playful" to gather user feedback on ease of use, ease of learning, naturalness and playfulness, respectively. We verbally asked to rank their preference and effort for each technique.

5 RESULTS

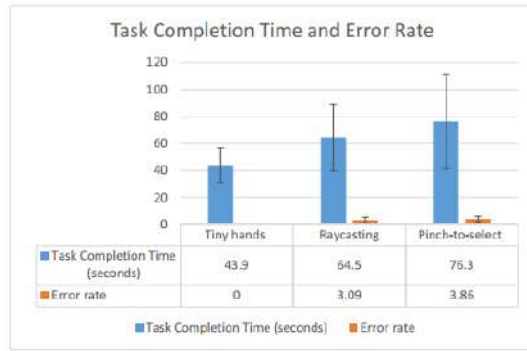
Using the Shapiro-Wilk test, we assessed the normality of the data. A one-way ANOVA test with a Bonferroni correction was conducted to find significant differences in normally distributed data. Additionally, we used the Friedman non-parametric test with Wilcoxon Signed Ranks post-hoc tests for Likert-scale data.

5.1 Task Completion Time

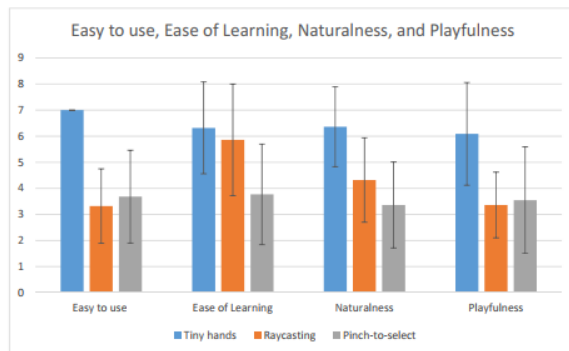
The study used ANOVA to analyze the differences in task completion time among the three techniques, with a significance level of 0.05. The results showed a statistically significant difference in task completion time between all three techniques ($F(2, 63)=8.86, p=.001$). Post hoc analysis using the Tukey test found that the tiny hands technique ($M=43.97, SD=12.90$) was significantly faster than both the pinch-to-select ($M=76.36, SD=34.93$) and ray casting ($M=64.51, SD=24.74$) techniques ($p<0.05$). There was no significant difference in task completion time between pinch-to-select and ray casting ($p>0.05$). Fig 3 presents the mean and SD values of task completion time for the three techniques.

5.2 Error Rate

To analyze the differences in error rates between the three techniques, we conducted ANOVA with a significance level of 0.05. The Tukey post hoc test indicated that there were significant differences in error rate between tiny hands ($M=0, SD=0$) and ray casting ($M=3.09, SD=2.1$), as well as between tiny hands and pinch-to-select ($M=3.86, SD=1.96, p<0.05$). Notably, the tiny hands technique produced zero errors.



(a)



(b)

Figure 3: (a) The mean and SD values of the task completion time and error rate for tiny hands, ray casting, and pinch-to-select and (b) The mean and SD error rate for tiny hands, ray casting and pinch-to-select.

5.3 Easy of use, Ease of learning, Naturalness, and Playfulness

A non-parametric Friedman test was conducted to compare the ease of use of all three techniques. The test showed a significant difference between the techniques, $\chi^2(2) = 28.93, p < 0.001$, indicating that at least one technique differed significantly from the others. Subsequently, post-hoc tests were performed using Wilcoxon Signed Ranks tests to identify the specific differences between the techniques. The results indicated that there were significant differences in the ease of use between tiny hands and ray casting ($Z=-3.91, p<0.05$), as well as between tiny hands and pinch-to-select ($Z=-4.01, p<0.05$) techniques. However, there was no significant difference between pinch-to-select and ray casting ($Z=-0.83, p<0.05$).

A significance test was conducted to compare the ease of learning among the techniques, yielding a $\chi^2(2) = 19.90, p < 0.001$, indicating a statistically significant result at $p<0.05$. Further analysis using the Wilcoxon Signed Ranks test revealed significant differences in the ease of learning between tiny hands and pinch-to-select ($Z=-1.81, p<0.05$), and between ray casting and pinch-to-select ($Z=-1.14, p<0.05$) techniques. No significant difference was found between tiny hands and ray casting ($Z=-0.13, p<0.05$).

The naturalness of the techniques was compared using a Friedman test, which yielded a significant difference between all the techniques ($\chi^2(2) = 22.90, p < 0.001$), indicating that there were significant differences in perceived naturalness among the techniques at a significance level of $p<0.05$. Wilcoxon Signed Ranks post-hoc tests were conducted to examine the differences between pairs of techniques. The results showed significant differences in naturalness between tiny hands and ray casting ($Z=-2.18, p<0.05$), between tiny hands and pinch-to-select ($Z=-3.21, p<0.05$), and between ray casting and pinch-to-select ($Z=-1.64, p<0.05$) techniques.

A statistically significant difference was observed in the playfulness measure across all the object selection techniques ($\chi^2(2) = 22.93, p < 0.001$) at a significance level of $p<0.05$. Wilcoxon Signed Ranks post-hoc tests were conducted to compare the techniques, which revealed a statistically significant difference in playfulness between tiny hands and ray casting ($Z=-2.63, p<0.05$). Similarly, a significant difference was found between tiny hands and pinch-to-select ($Z=-2.32, p<0.05$). We also observed a statistically significant difference between ray casting and pinch-to-select ($Z=-0.84, p<0.05$).

5.4 Preference and Effort

All 22 participants indicated a clear preference for the tiny hands technique as their first preference over the other two techniques. No participants indicated tiny hands for the second and third preference. Ray casting was ranked as the second preference by 15 out of 22 participants. 7 out of 22 ranked ray casting as their third preference. Only 7 participants ranked pinch-to-select as their second preference. 15 participants indicated pinch-to-select as their third preference.

Participants perceived the tiny hands technique as requiring less effort compared to the other techniques (20 out of 22). Among the three techniques, the pinch-to-select technique was rated as more effortful by participants (18 out of 22).

6 DISCUSSION

Our results clearly show that tiny hands was significantly faster in terms of task completion time and error rate as compared to other techniques. The participants reported that the technique's effectiveness could be attributed to two key factors: firstly, the ability to select small objects by making the virtual hands small, and secondly, the ease of navigating within a dense VE due to the small size of the virtual hands. A participant stated that "I found that the smaller size of the virtual hands is easy to map to the similar size objects, hence giving me the confidence to select them accurately." Additionally, the small size of the virtual hands allows for a smaller selection volume, resulting in higher accuracy and reduced errors when using the tiny hands technique. One participant mentioned that "I like the fact that the smaller size of the virtual hands allowed for more precise movements during my selections compared to other techniques." These findings suggest that the tiny hands technique could help select smaller objects in dense VEs, particularly when precision is required. Similar findings were reported in [11], where a scaled-down version of a life-size avatar took significantly less time to complete a task than a life-size avatar. The scaled-down version was also less fatiguing and led to a low mental load in the study [11]. With ray casting, participants had to precisely point towards a small object, often resulting in higher selection time. Pointing to select a small target using ray-casting also led to incorrect selection and inaccuracies due to hand jitter and difficulty in accurately pointing the ray as the VE was dense – a phenomenon commonly observed in the ray-casting technique [2]. With the pinch-to-select technique, we found that participants experienced difficulties in accurately selecting small objects. One of the reasons for this could be the pinch gesture's limited precision, as participants could not correctly measure the distance between their thumb and index finger while selecting the object. This might have led to incorrect object selection. Our findings are consistent with previous studies that reported limited precision with the pinch-to-select technique, which resulted in difficulties in accurately selecting small objects [7].

Tiny hands also had significantly increased ease of use and learning compared to ray-casting and pinch-to-select. This was attributed to its ability to customize the technique for a greater sense of control and confidence. The use of the non-dominant hand to trigger the tiny hands and further customize it by vertical movements was well-received by the participants, as it provided a sense of control and confidence among the participants. One participant stated, "Using vertical movement to customize the size of the virtual hand felt natural and intuitive and gave me a sense of control and confidence that I could easily navigate and select the desired objects in the VE." The difficulties encountered in selecting small objects with the ray casting and pinch-to-select techniques were mainly due to their lack of customization options, which led to a poor sense of control and confidence [9], preventing users from adapting them their specific needs and the constraints of the VE. In contrast, the tiny hands technique allowed for greater customization and adaptability, which resulted in participants feeling more in control and confident while using it. This resulted in significantly higher ease of use and learning for tiny hands than the ray casting and pinch-to-select technique.

Tiny hands also had significantly increased naturalness. This was primarily due to two factors: First, the technique's mental model closely resembled traditional graphical user interfaces' mouse and cursor-based selection. This familiarity made it easy for participants to understand and use the technique. One participant stated, "It feels like using a mouse and cursor that we have used for years. The dominant hand feels like the mouse, which is controlling the tiny hands that act as a cursor." Second, the significantly better naturalness of tiny hands can be attributed to its direct selection approach for object selection and appropriate mapping of small virtual hands to select small virtual objects. It made the technique intuitive for interacting with dense VEs with small objects. This finding aligns with previous studies demonstrating the effectiveness of direct selection and manipulation techniques in increasing naturalness in VR [14]. In contrast, participants found ray casting and pinch-to-select less natural as these techniques adopted an indirect selection approach and adopted hand as a controller, which felt less intuitive and less in line with the natural movement of their hands. Pinch-to-select, in particular, was challenging for participants as they often associated the pinch gesture with zooming or scaling objects, as is common on mobile devices. One participant commented, "I found it difficult to select objects using pinch-to-select because my brain kept associating the pinch gesture with zooming or scaling as I do with images on my phone."

During our study, we observed that participants did not typically set their tiny hand size to the smallest value available, which was 0.1. Instead, participants typically selected a range between 0.2 and 0.5 as their preferred tiny hand size. When asked about their reasoning, participants explained that setting the hand size too small made it challenging to control and perceive the distance between the hand and the objects in the VE. One participant mentioned, "I feel like if it's too small, I might not be able to control it properly, and it seems like it's too far away from the objects." Another participant mentioned that they preferred to compare the target size with the hand size to ensure accurate selection but did not feel it necessary to make the tiny hands extremely small. He commented, "I am comparing the target size with the hand size and confirming the tiny hand size. I feel it need not be very small to select the object accurately."

A few participants reported a few challenges in using tiny hands. They felt that using the tiny hands required more movement than other techniques. They also noted that moving the virtual hand away from the user made the tiny hands appear smaller, resulting in a feeling of distance. We believe this is due to depth perception – a phenomenon in which humans underestimate the distance of distant visual objects and overestimate the distance of near visual objects [4]. A participant commented, "It feels like I have to move more to move the tiny hands. Moving the real hand away from me makes the tiny hands look smaller. Hence it feels like it's far away." A few participants suggested the need for a visualization, such as a scale, to indicate how much to vertically move to decrease or increase virtual hand size. One participant commented, "A visualization in the form of a scale could help me understand how much to decrease and increase the hand size. I also want to understand the current hand size that I am using while using tiny hands."

7 CONCLUSION AND FUTURE WORK

In this study, we investigated the efficacy of the tiny hands object selection technique in selecting small objects within arm's reach in a dense VE relative to the ray casting and pinch-to-select techniques. The tiny hands technique involves reducing the size of virtual hands to enable a more accurate selection of small objects. A comparative study was conducted, and the results demonstrated that the tiny hands technique outperformed the ray casting and pinch-to-select techniques regarding selection speed and accuracy. Furthermore, participants found the tiny hands technique to be more playful, natural, and easy to use and learn, primarily due to the ability to select small objects by resizing the virtual hands and the ease of navigating a dense VE using small virtual hands. Participants also appreciated the similarity of the tiny hands technique to the traditional mouse and cursor-based interactions, as well as the flexibility and user control in adjusting the hand size for different object sizes. The tiny hands technique required additional movements, which can lead to fatigue with prolonged use, and participants reported a sense of incorrect depth perception. Overall, the findings suggest that the tiny hands technique has the potential to be a viable alternative for the accurate and efficient selection of small objects within arm's reach in dense VEs. We believe further research is necessary to address its limitations and determine its potential for wider adoption.

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